



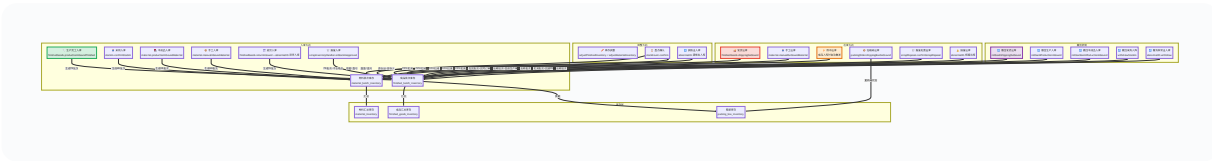
仓储管理 — 全流程方案

基于 Seals MES System 现有代码分析 | 成品仓 + 物料仓 + 采购入库 + 盘点 + 装箱 + 报废处置
+ 其他出入库

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一、全局流程总览



核心服务: `server/src/services/warehouse/finishedGoods.ts` + `material.ts` | 路由聚合:
`server/src/modules/warehouse/index.ts`

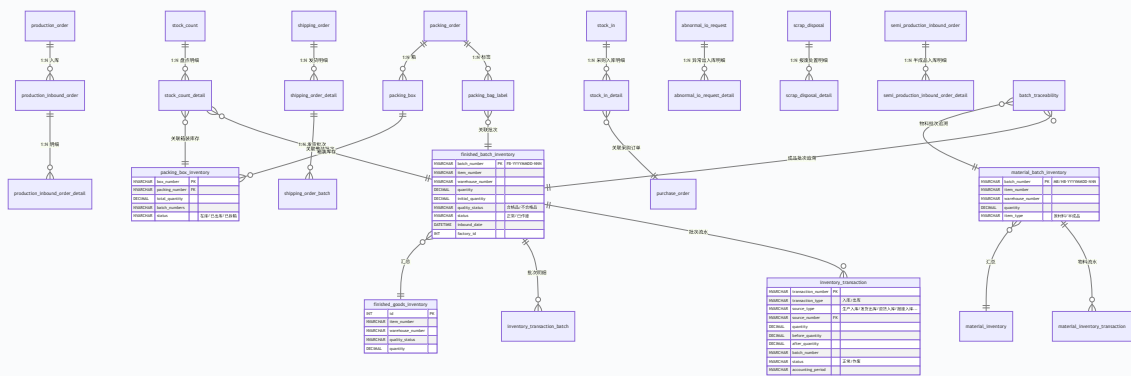
二、各单据编号规则

单据	编号格式	前缀	示例	来源模块
生产 入库 单	PI-YYYYMMDD-NNN	PI	PI-20260601-001	finishedGoods (自动生成)
成品 批次 号	FB-YYYYMMDD-NNN	FB	FB-20260601-001	inventory.service (generateBatchNumber)
原材 料批 次号	MB-YYYYMMDD-NNN	MB	MB-20260601-001	inventory.service
半成 品批 次号	HB-YYYYMMDD-NNN	HB	HB-20260601-001	inventory.service
采购 入库 单	SI {FC} -YYYYMMDD-NNN	SI	SI-20260601-001	stockIn.controller

发货单	SO-YYYYMMDD-NNN	SO	SO-20260601-001	finishedGoods (自动生成)
盘点单	IC {FC} -YYYYMMDD-NNN	IC	IC-20260601-001	stockCount.controller
装箱单	PK {FC} YYYYMMDDNNN	PK	PK20260601001	packingOrder.controller
箱号	BOXYYYYMMDDNNN	BOX	BOX20260601001	packingOrder (自动生成)
报废处置单	SD {FC} -YYYYMMDD-NNN	SD	SD-20260601-001	scrapDisposal.controller
退货入库单(异常出入库)	RI-YYYYMMDD-NNN	RI	RI-20260601-001	abnormalIO.controller
报废出库单(异常出入库)	SO-YYYYMMDD-NNN	SO	SO-20260601-001	abnormalIO.controller
调拨出入库单	TR-YYYYMMDD-NNN	TR	TR-20260601-001	abnormalIO.controller
盘盈盘亏单	SC-YYYYMMDD-NNN	SC	SC-20260601-001	abnormalIO.controller

注: {FC} 为工厂代码 (多工厂场景), 批次号通过 generateBatchNumber() 自动生成并保证全局唯一。

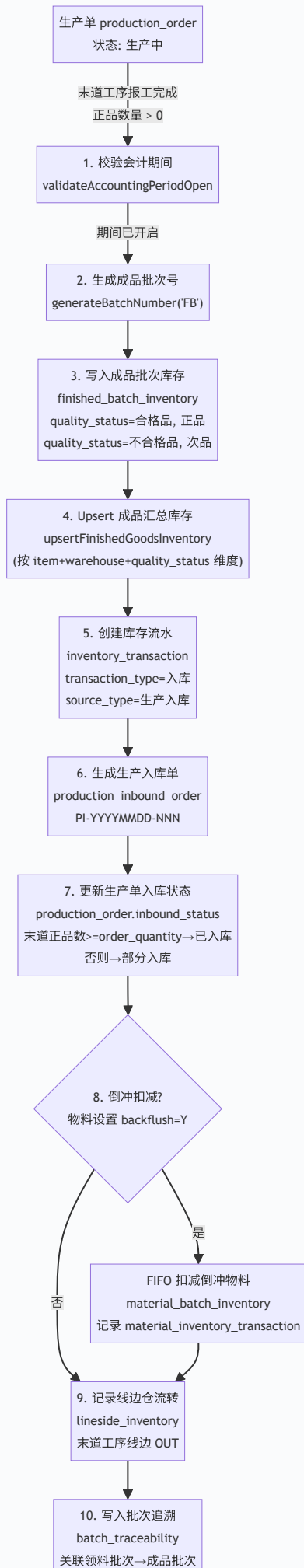
三、仓储数据模型 (ER 图)



数据库表定义分散在各 SQL 查询中，核心库存表: finished_batch_inventory, material_batch_inventory, finished_goods_inventory, material_inventory, inventory_transaction, inventory_transaction_batch

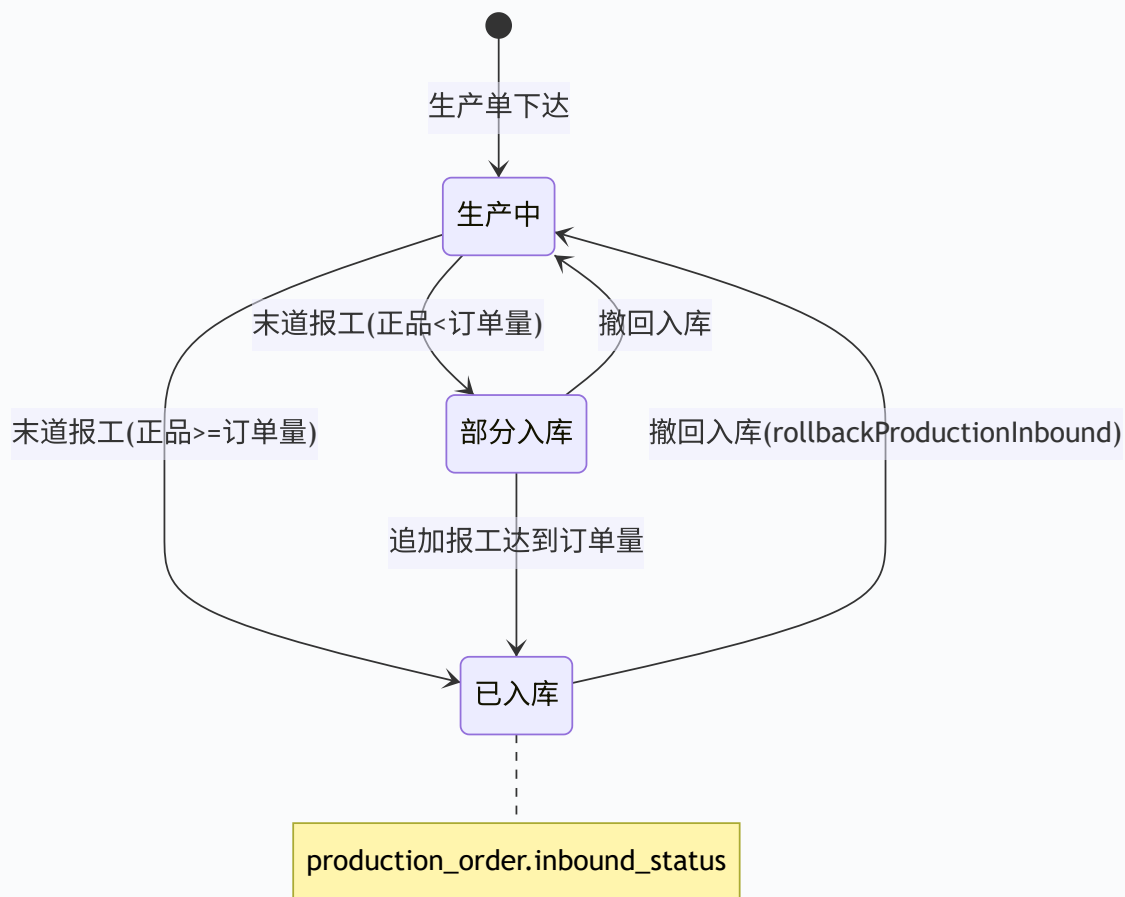
四、成品仓 — 生产入库流程

4.1 流程图



✓ 入库完成

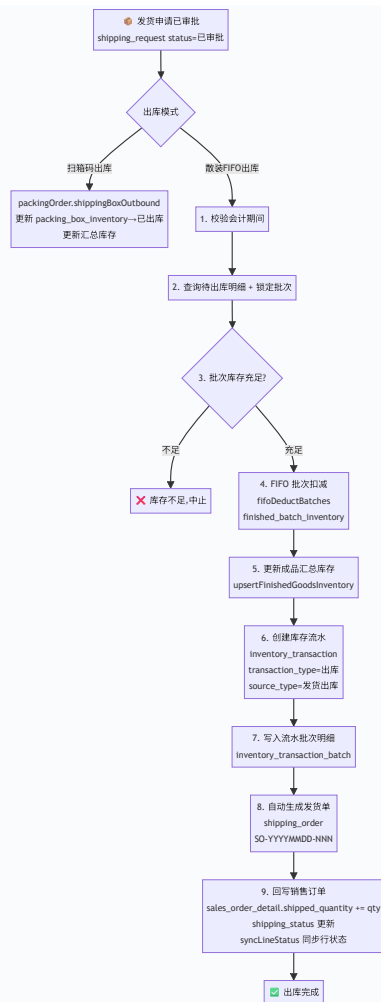
4.2 状态机



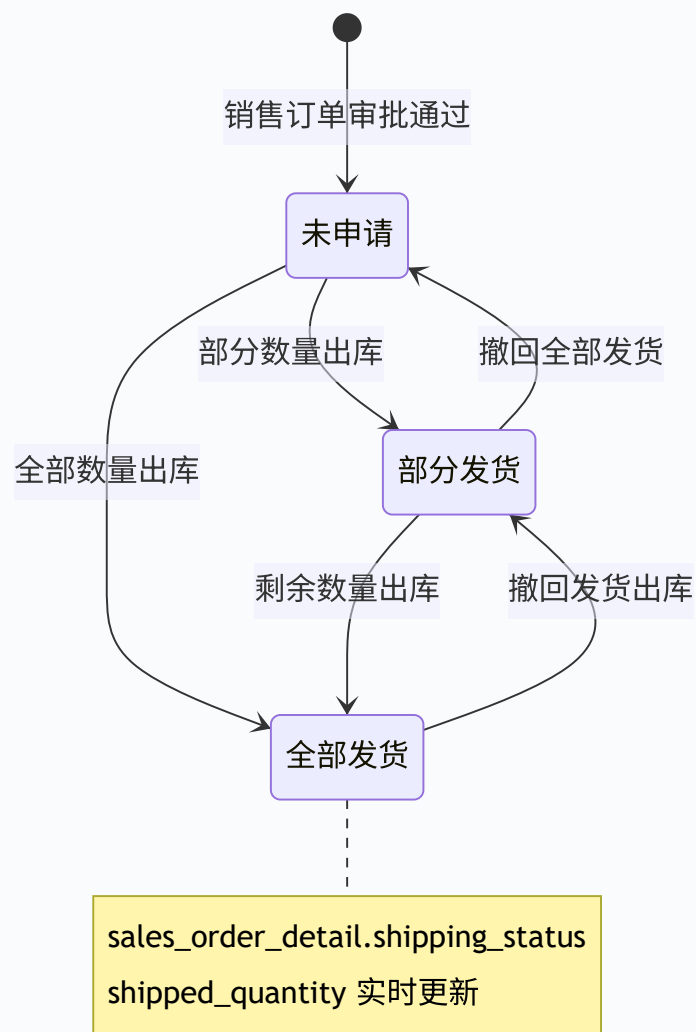
server/src/services/warehouse/finishedGoods.ts → productionInboundFinished() (L27-217)

五、成品仓 — 发货出库流程

5.1 流程图



5.2 状态机



server/src/services/warehouse/finishedGoods.ts → shippingOutbound() (L221-527) | 扫描码出库:
packingOrder.controller.ts → shippingBoxOutbound() (L814-946)

六、成品仓 — 退货入库流程



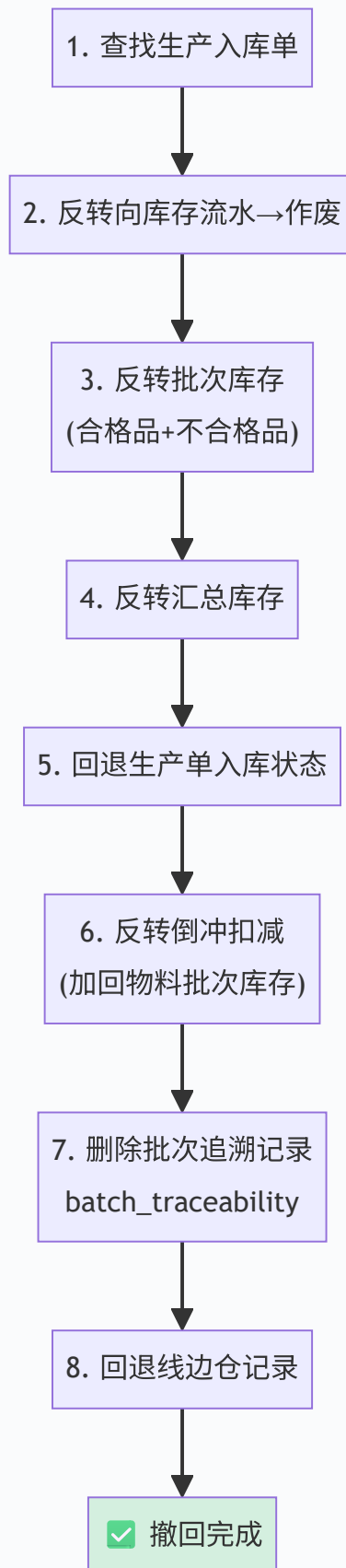
server/src/services/warehouse/finishedGoods.ts → returnInbound() (L531-678) | 另有异常出入库退货: abnormalIO.controller.ts → handleReturnInbound() (L470-560)

七、成品仓 — 撤回与回滚机制

7.1 撤回发货出库 (rollbackShippingOutbound)

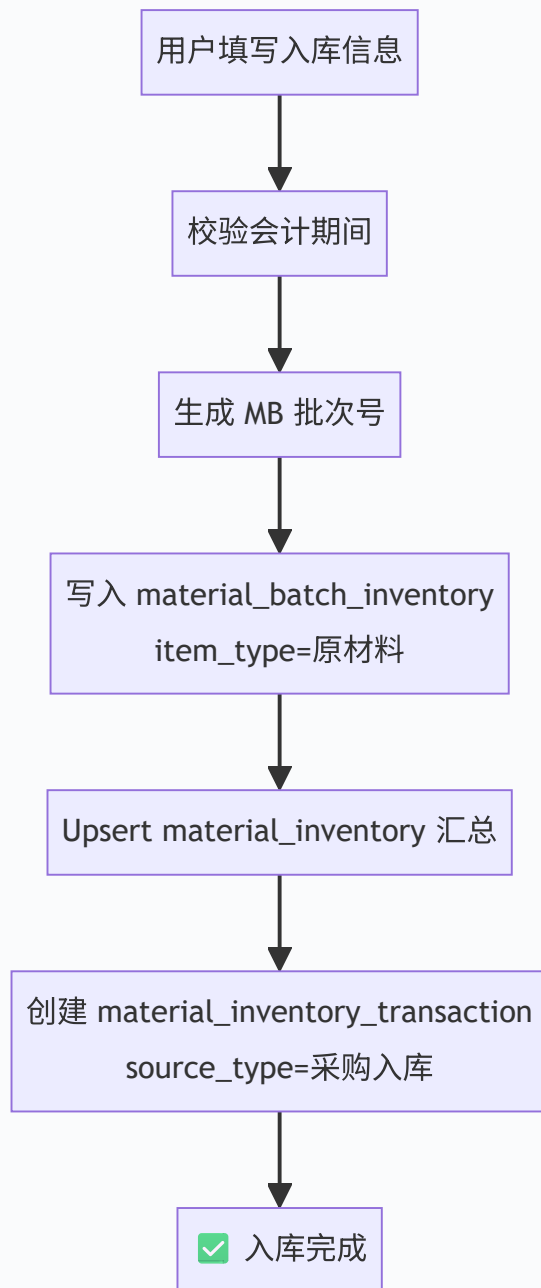


7.2 撤回生产入库 (rollbackProductionInbound)



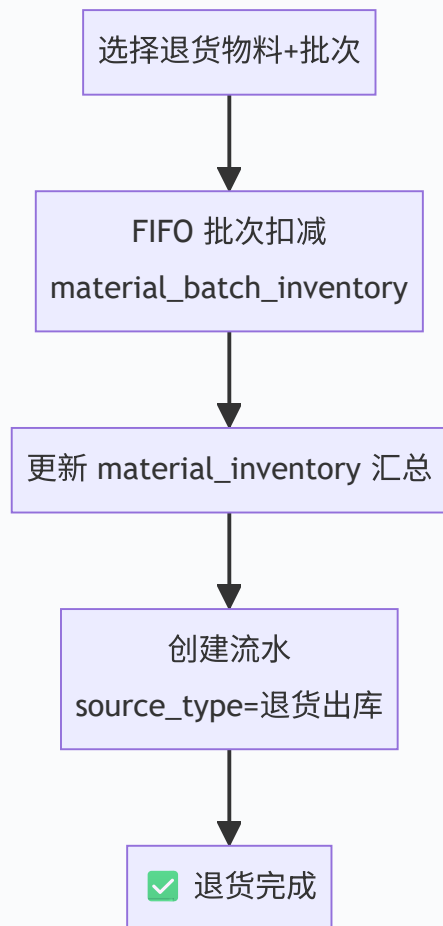
八、物料仓 — 原材料入库流程

8.1 手工入库



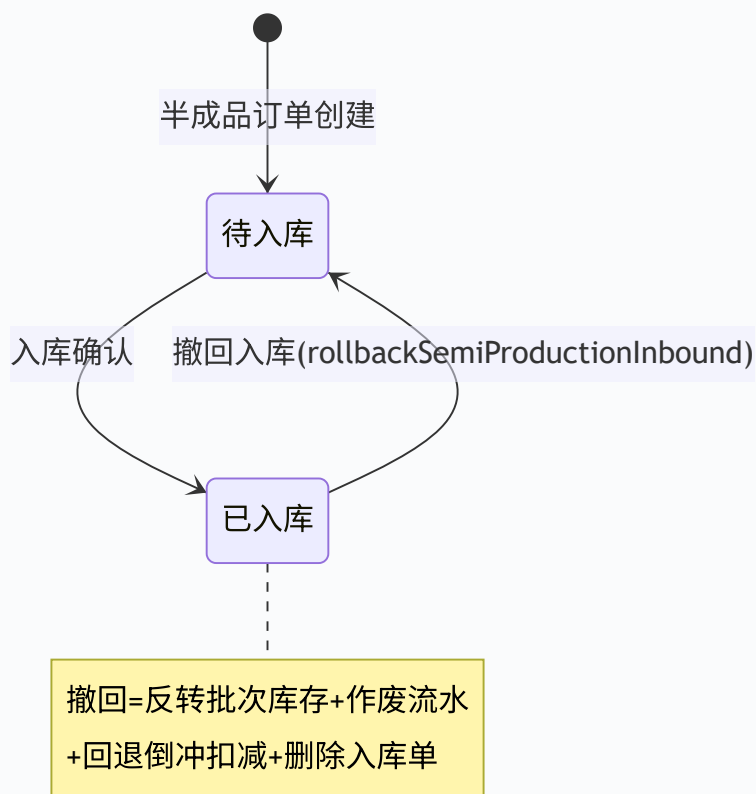
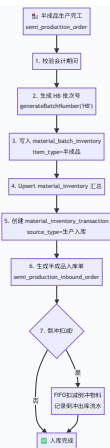
📁 `material.ts` → `manualInboundMaterial()` (L20-107)

8.2 采购退货出库



materialWarehouse.controller.ts → executeReturnOutbound()

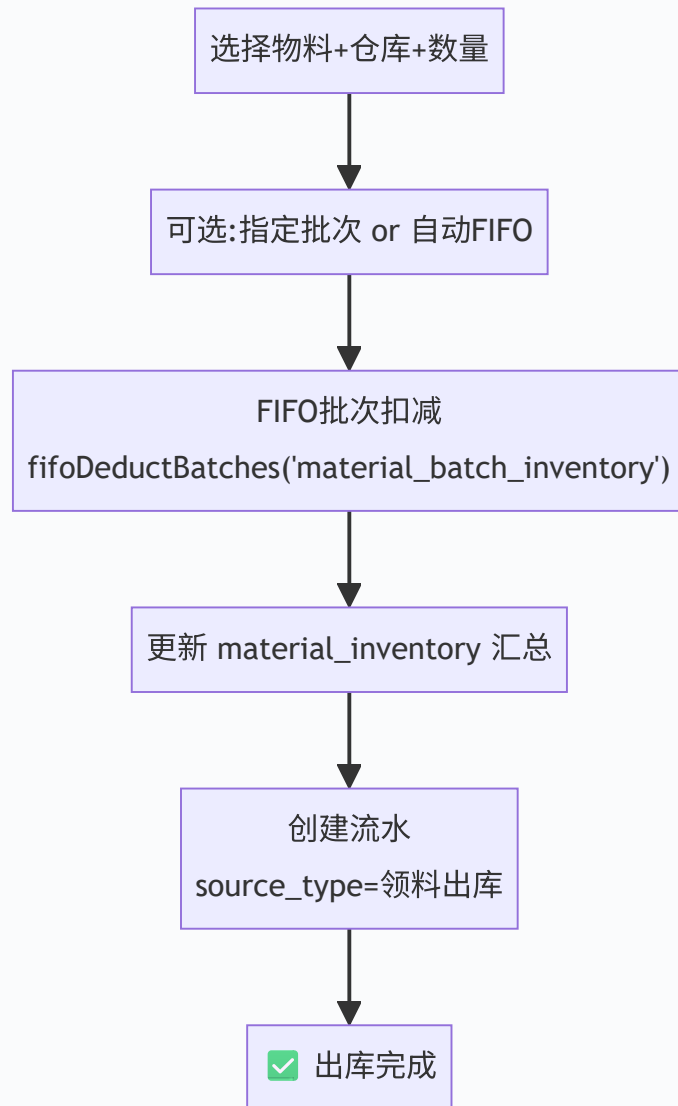
九、物料仓 — 半成品入库流程



server/src/services/warehouse/material.ts → productionInboundMaterial() (L110-270) |
 rollbackSemiProductionInbound() (L448-631)

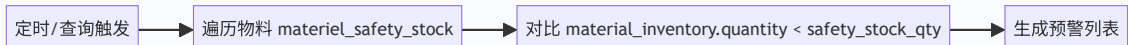
十、物料仓 — 出库与撤回

10.1 手工出库 (FIFO)



material.ts → manualOutboundMaterial() (L274-364)

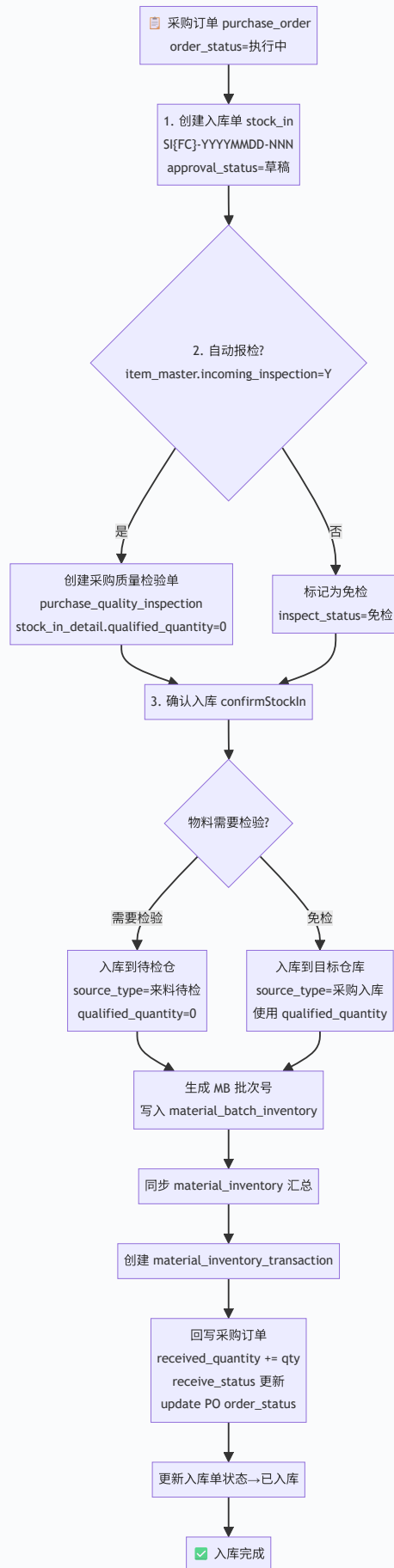
10.2 安全库存预警



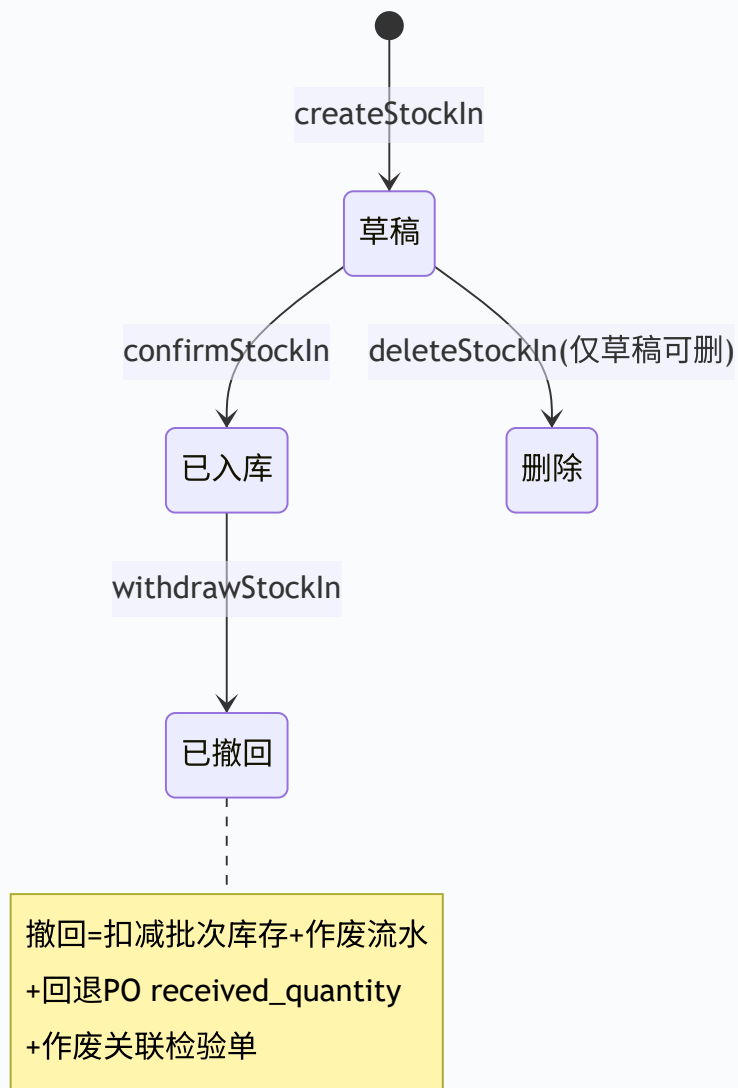
materialWarehouse.controller.ts → getSafetyStockAlerts()

十一、采购入库 (stock_in) 全流程

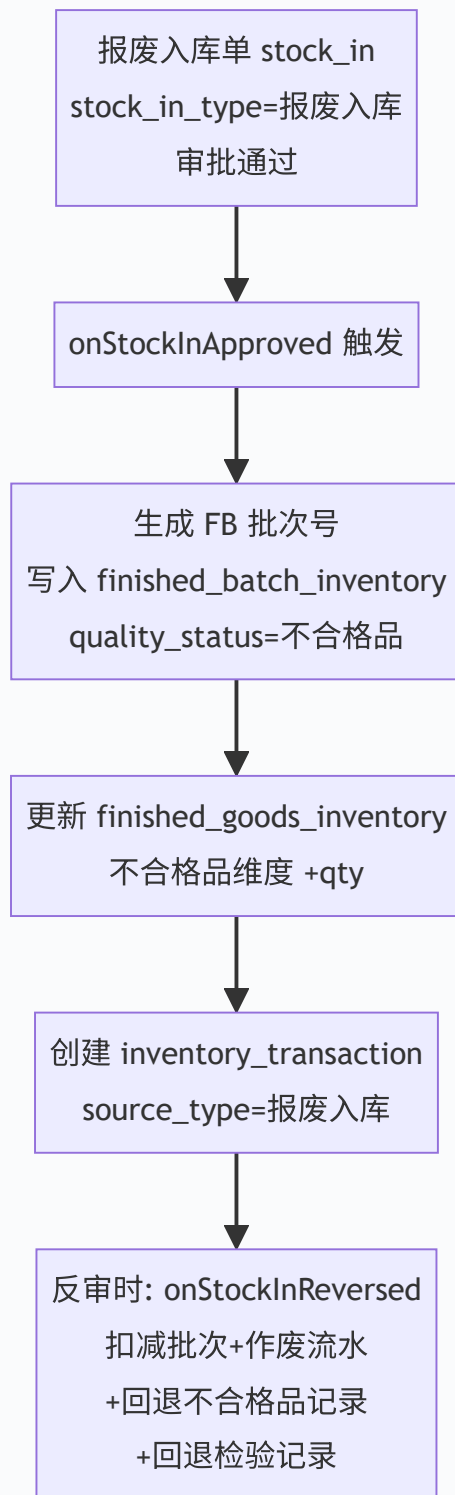
11.1 流程图



11.2 状态机



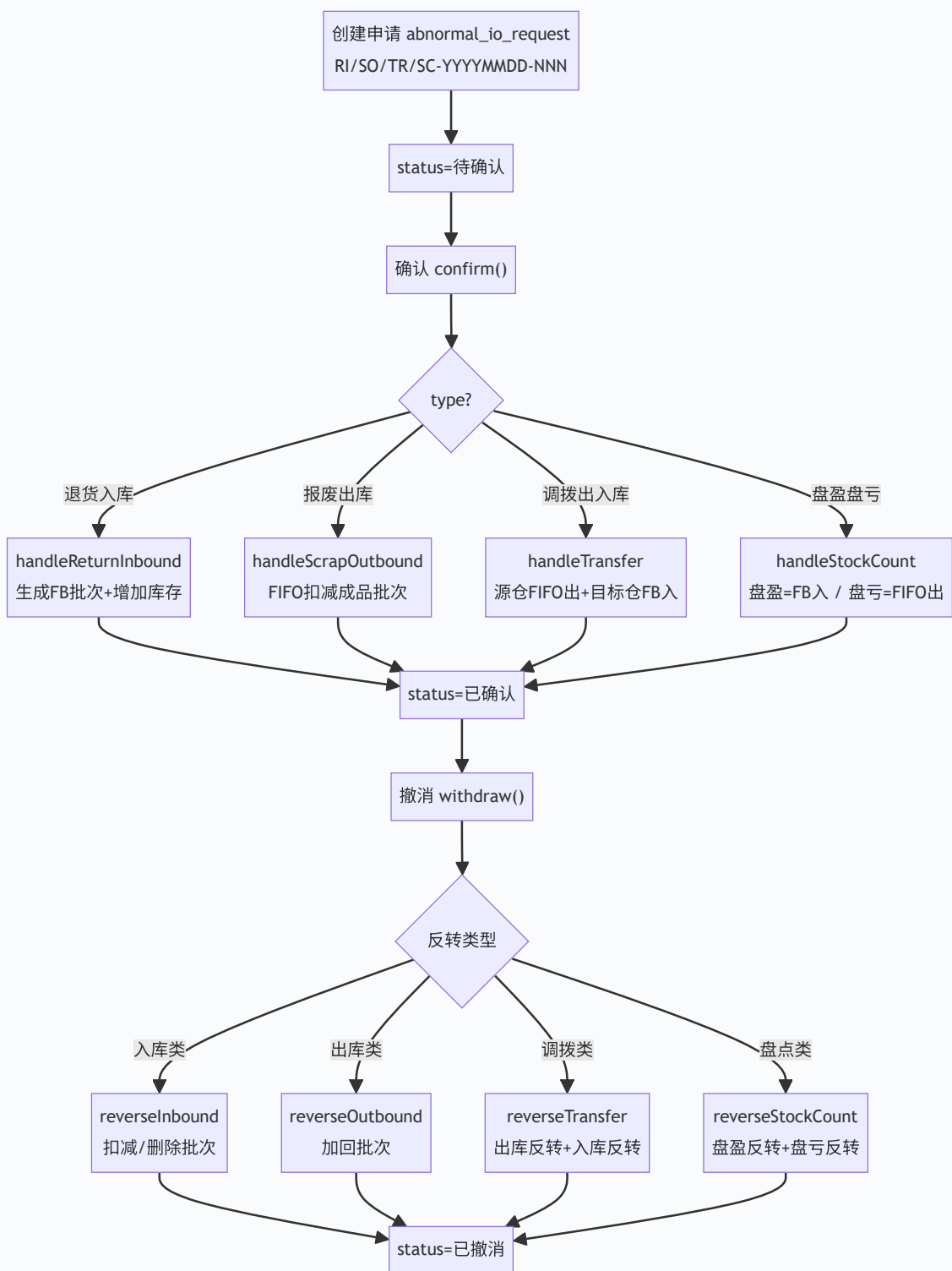
11.3 报废入库处理



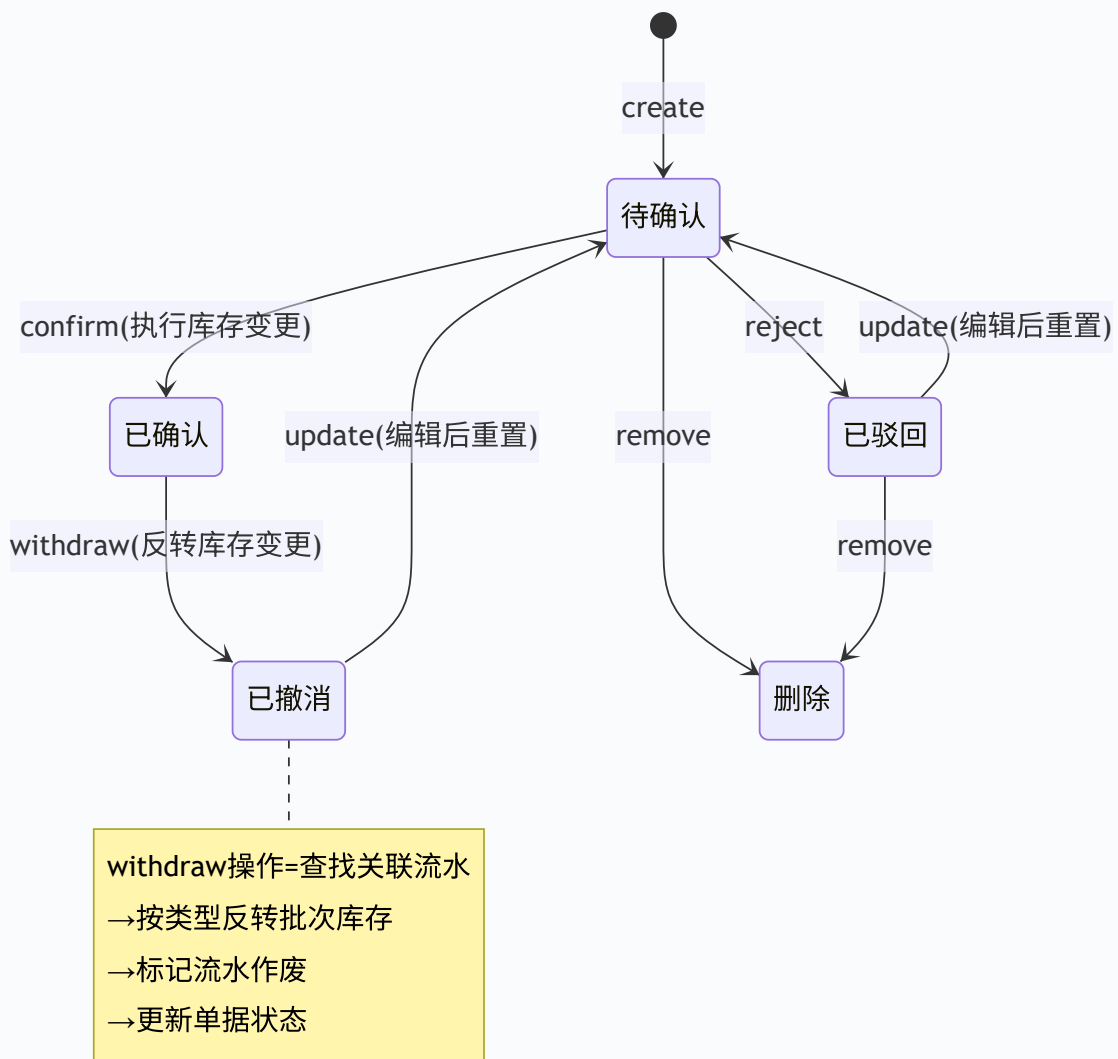
server/src/modules/warehouse/stockIn/stockIn.controller.ts (865行) | scrapInventoryHandler.ts (343行)

十二、其他出入库 (abnormal_io) 全流程

12.1 四种类型总览



12.2 状态机



server/src/modules/warehouse/abnormalIO/abnormalIO.controller.ts (1172行)

十三、盘点管理 (stock_count) 全流程

13.1 流程图



Syntax error in text
mermaid version 10.9.6

13.2 状态机

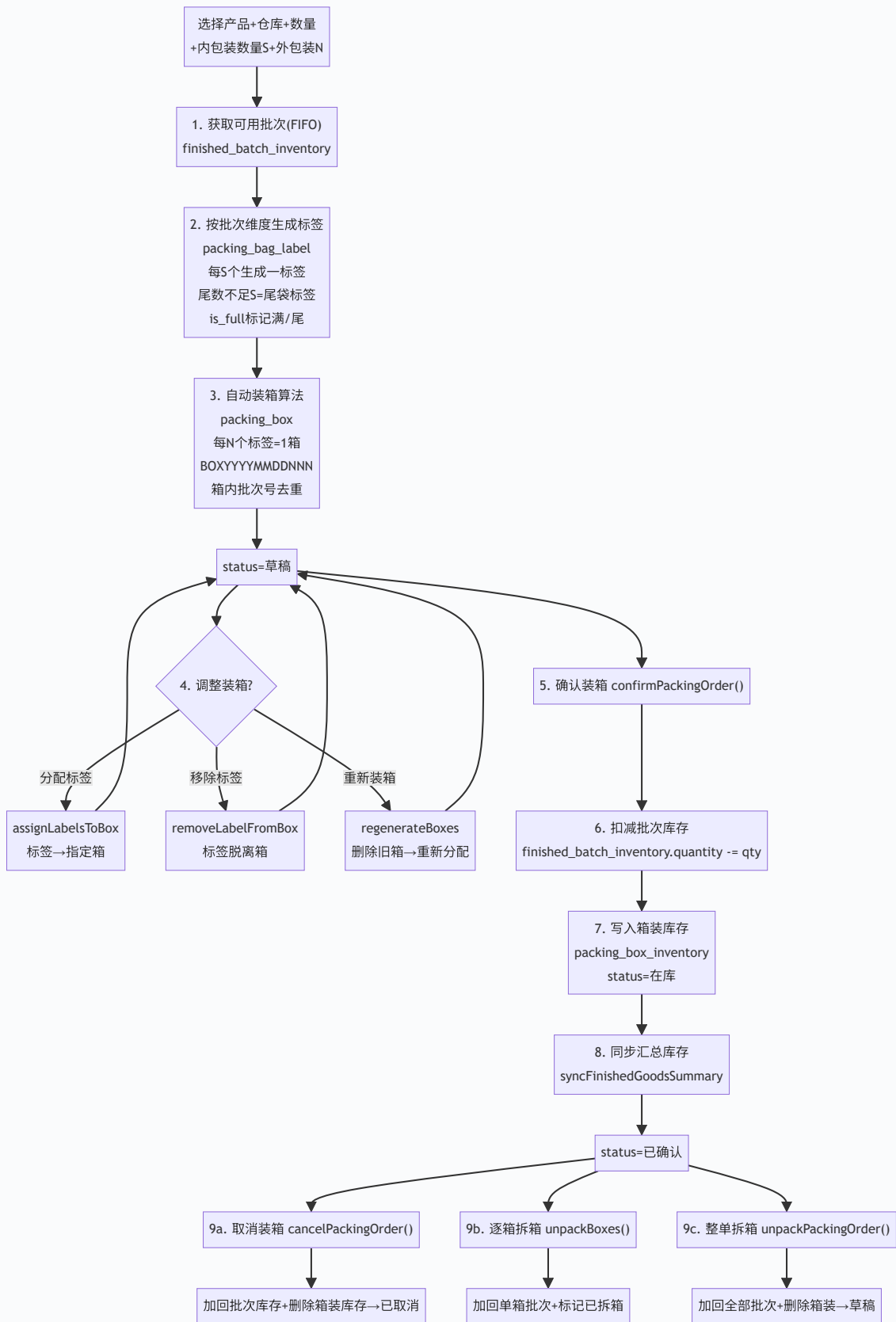


Syntax error in text
mermaid version 10.9.6

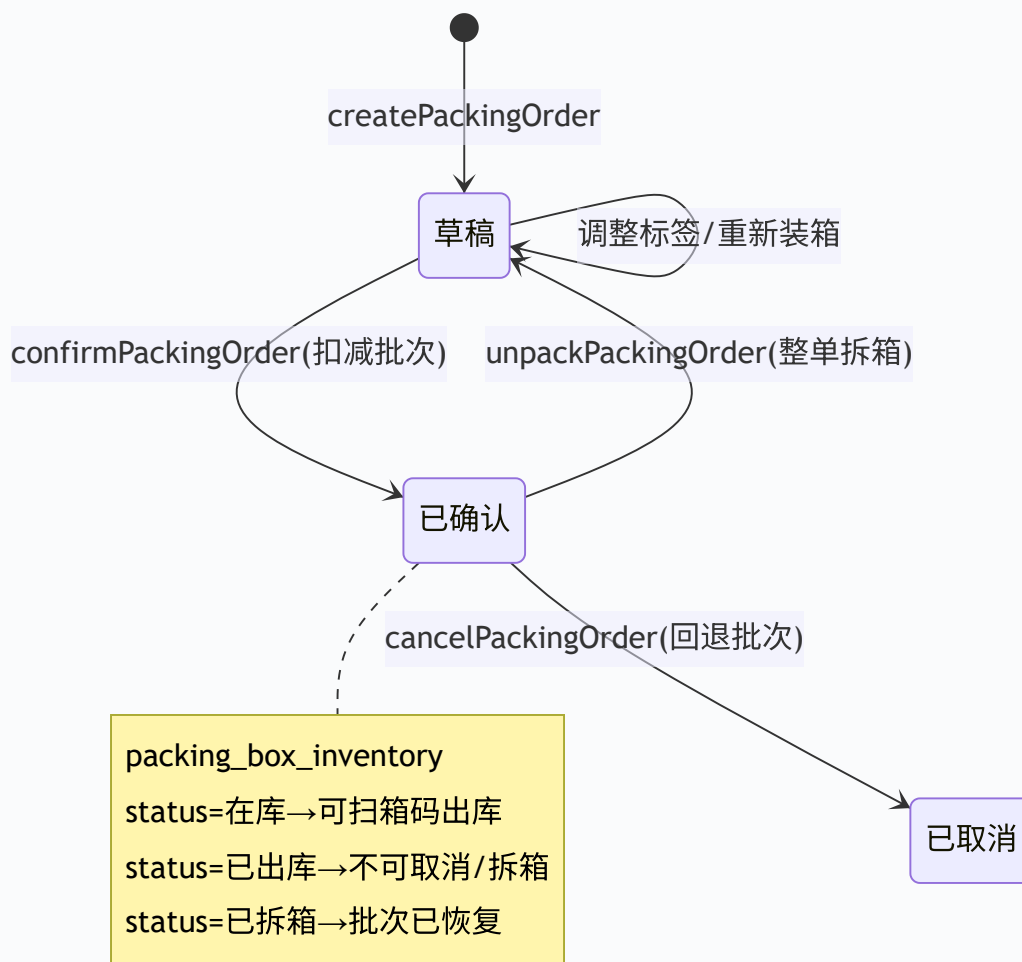
server/src/modules/warehouse/stockCount/stockCount.controller.ts (1269行)

十四、装箱单管理 (packing_order) 全流程

14.1 流程图

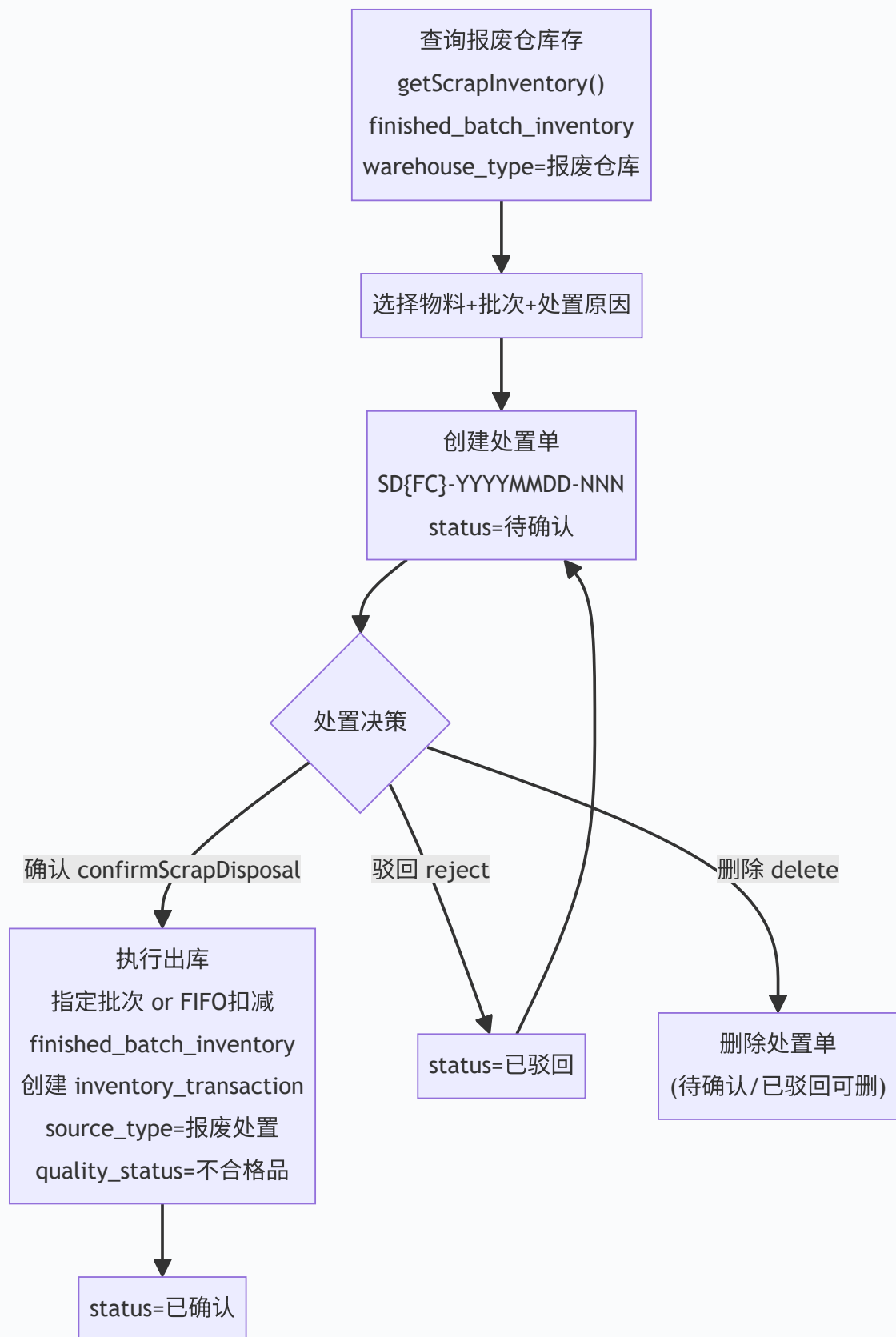


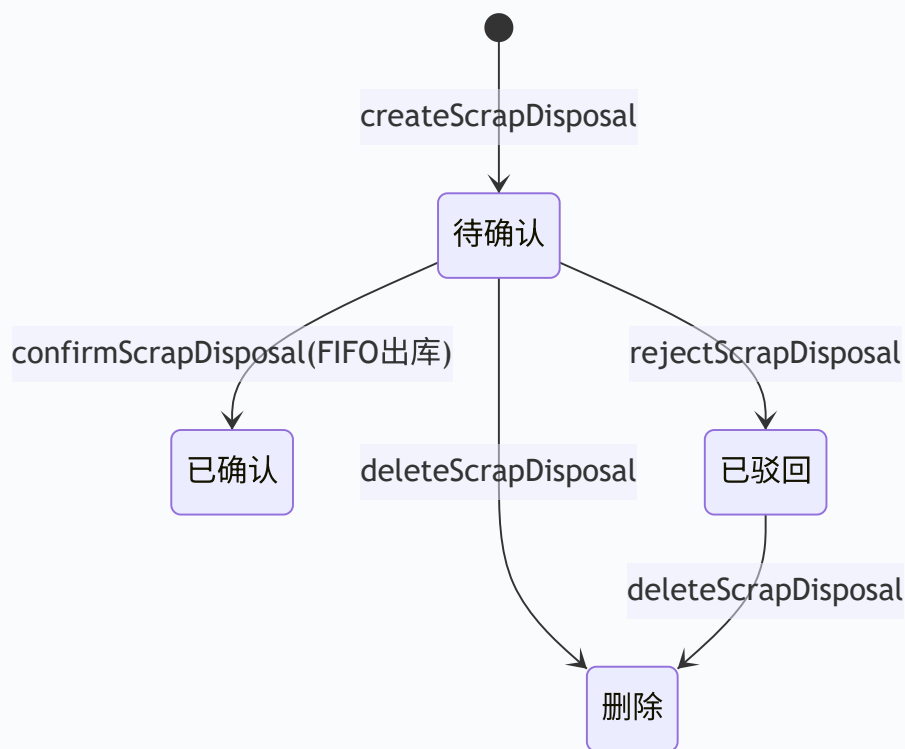
14.2 状态机



server/src/modules/warehouse/packingOrder/packingOrder.controller.ts (1165行)

十五、报废处置管理 (scrap_disposal)



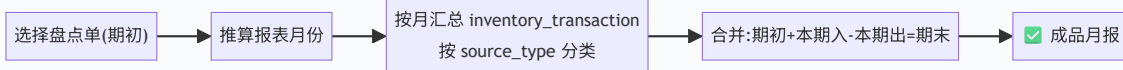


server/src/modules/warehouse/scrapDisposal/scrapDisposal.controller.ts (618行) | 月度报表:
scrapMonthlyReport.controller.ts (165行)

十六、月度报表体系

16.1 成品月度出入库报表

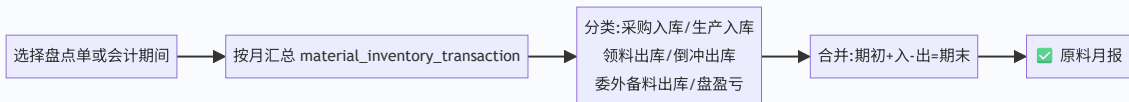
基于 `inventory_transaction` + 盘点单(`stock_count`)期初数据, 按会计期间汇总。



finishedGoods.controller.ts → getMonthlyReport()

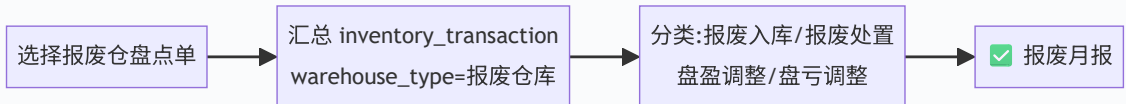
16.2 原料仓月度报表

基于 `material_inventory_transaction` + 盘点单期初。



📁 `materialMonthlyReport.controller.ts` (369行) → `getMaterialMonthlyReport()` + `getMaterialMonthlyReportByPeriod()`

16.3 报废仓月度报表



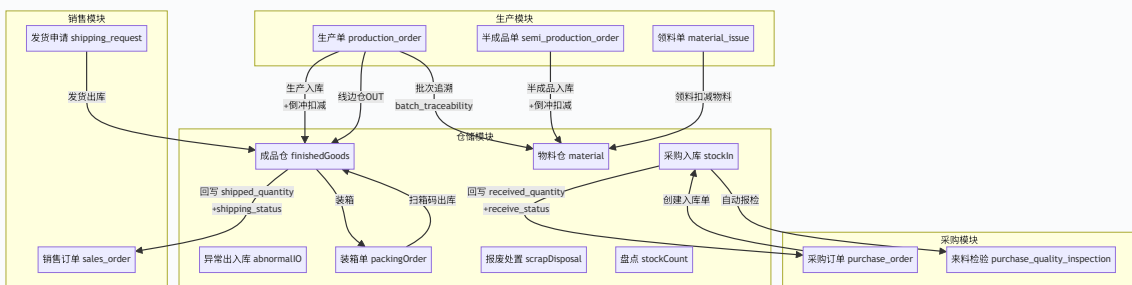
📁 `scrapMonthlyReport.controller.ts` (165行)

16.4 有效期管理报告



📁 `shelfLifeReport.controller.ts` (110行)

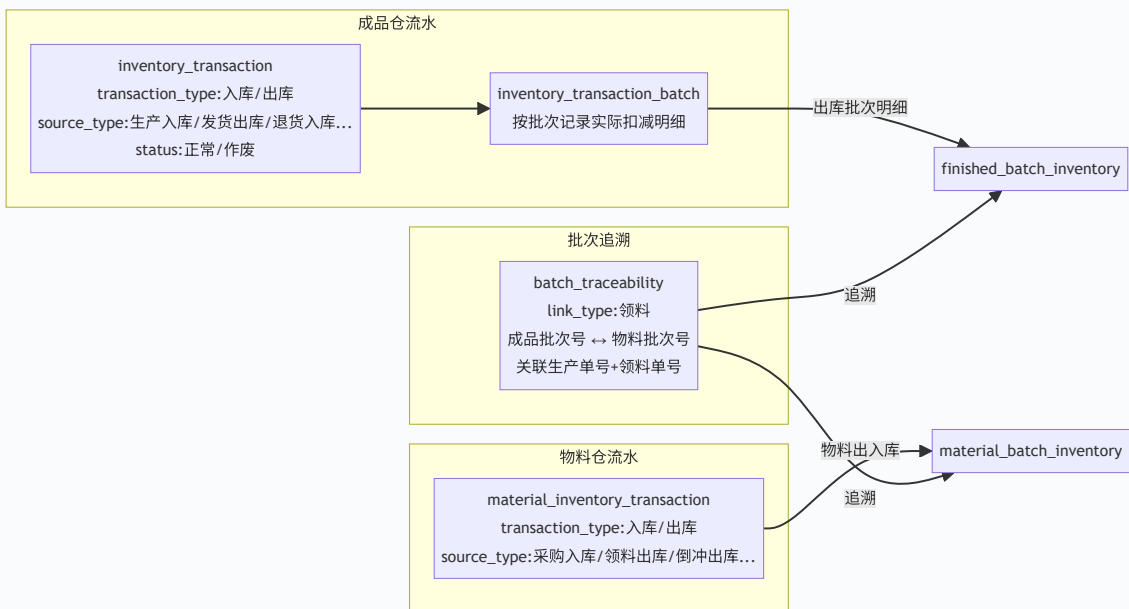
十七、跨模块联动图



关键联动 生产入库→倒冲扣减物料仓 | 发货出库→回写销售订单 | 采购入库→回写采购订单+自动报检 | 生产入库→批次追溯关联领料批次

十八、库存流水与批次追溯

18.1 库存流水层级



18.2 FIFO 先进先出策略

所有出库操作均按 `inbound_date ASC, id ASC` 排序扣减批次库存。指定批次时直接扣减指定批次。函数 `fifoDeductBatches()` 统一处理成品和物料批次扣减。

`server/src/services/warehouse/helpers.ts` → `fifoDeductBatches()` (L128-190)

十九、API 清单

19.1 成品仓 (finished-goods)

方法	路径	说明	文件位置
GET	<code>/api/finished-goods/inventory</code>	成品+原材料库存列表 (UNION)	<code>finishedGoods.controller.ts</code>
GET	<code>/api/finished-goods/inventory/:item_number/:warehouse_number</code>	库存详情 (含流水)	<code>finishedGoods.controller.ts</code>
GET	<code>/api/finished-goods/pending-inbound</code>	待入库生产单	<code>finishedGoods.controller.ts</code>
GET	<code>/api/finished-goods/pending-outbound</code>	待出库发货申请	<code>finishedGoods.controller.ts</code>
GET	<code>/api/finished-goods/pending-return-inbound</code>	待退货入库	<code>finishedGoods.controller.ts</code>
POST	<code>/api/finished-goods/production-inbound</code>	生产完工入库	<code>finishedGoods.controller.ts</code>
POST	<code>/api/finished-goods/shipping-outbound</code>	发货出库	<code>finishedGoods.controller.ts</code>
POST	<code>/api/finished-goods/return-inbound</code>	退货入库	<code>finishedGoods.controller.ts</code>
POST	<code>/api/finished-goods/adjust-inventory</code>	库存调整	<code>finishedGoods.controller.ts</code>

POST	/api/finished-goods/rollback-outbound	撤回发货 出库	finishedGoods.controller.ts
POST	/api/finished-goods/withdraw-inbound-order	撤回生产 入库单	finishedGoods.controller.ts
GET	/api/finished-goods/monthly-report	成品月度 出入库报 表	finishedGoods.controller.ts

19.2 物料仓 (material-warehouse)

方法	路径	说明
GET	/api/material-warehouse/inventory	物料库存列表
GET	/api/material-warehouse/pending-inbound	待入库半成品
GET	/api/material-warehouse/batch-options	批量FIFO推荐批次
GET	/api/material-warehouse/safety-stock-alerts	安全库存预警
GET	/api/material-warehouse/return-outbound	采购退货出库列表
POST	/api/material-warehouse/manual-inbound	手工入库
POST	/api/material-warehouse/production-inbound	半成品生产入库
POST	/api/material-warehouse/manual-outbound	手工出库
POST	/api/material-warehouse/adjust-inventory	库存调整
POST	/api/material-warehouse/withdraw-semi-inbound	撤回半成品入库
POST	/api/material-warehouse/execute-return-outbound	执行退货出库
GET	/api/material-warehouse/monthly-report	原料仓月度报表

19.3 采购入库 (stock-ins)

方法	路径	说明
GET	/api/stock-ins	入库单列表

GET	/api/stock-ins/:id	入库单详情
POST	/api/stock-ins	创建入库单(含自动报检)
DELETE	/api/stock-ins/:id	删除入库单(仅草稿)
POST	/api/stock-ins/:id/confirm	确认入库(执行库存变更)
POST	/api/stock-ins/:id/withdraw	撤回入库
GET	/api/stock-ins/export	导出Excel
GET	/api/stock-ins/purchase-reconciliation	采购对账分页
PUT	/api/stock-ins/purchase-reconciliation/status	更新对账状态

19.4 盘点 (stock-counts)

方法	路径	说明
GET	/api/stock-counts	盘点单列表(含状态统计)
GET	/api/stock-counts/snapshot-preview	快照预览
GET	/api/stock-counts/:count_number	盘点单详情
POST	/api/stock-counts	创建盘点单(快照库存)
PUT	/api/stock-counts/:count_number	录入实盘数量
DELETE	/api/stock-counts/:count_number	删除(仅盘点中)
POST	/api/stock-counts/:count_number/submit-review	提交复核
POST	/api/stock-counts/:count_number/review	复核(通过/驳回)
POST	/api/stock-counts/:count_number/confirm	确认执行(库存调整)
POST	/api/stock-counts/:count_number/cancel	作废
GET	/api/stock-counts/report/summary	盘点汇总报表
GET	/api/stock-counts/report/diff-detail	差异明细报表
GET	/api/stock-counts/report/trend	盘点趋势报表

19.5 异常出入库 (abnormal-io)

方法	路径	说明
GET	/api/abnormal-io	列表(按type/status筛选)
GET	/api/abnormal-io/:request_number	详情
POST	/api/abnormal-io	创建(退货入库/报废出库/调拨/盘点)
PUT	/api/abnormal-io/:request_number	更新(待确认/已驳回/已撤消可编辑)
DELETE	/api/abnormal-io/:request_number	删除
POST	/api/abnormal-io/:request_number/confirm	确认(执行库存变更)
POST	/api/abnormal-io/:request_number/reject	驳回
POST	/api/abnormal-io/:request_number/withdraw	撤消(反转库存变更)

19.6 装箱单 (packing-orders)

方法	路径	说明
GET	/api/packing-orders	装箱单列表
GET	/api/packing-orders/:packingNumber	详情(含标签+箱)
GET	/api/packing-orders/available-batches	可用批次
POST	/api/packing-orders	创建(含标签+自动装箱)
POST	/api/packing-orders/:packingNumber/confirm	确认(扣减批次+箱装库存)
POST	/api/packing-orders/:packingNumber/cancel	取消
POST	/api/packing-orders/:packingNumber/assign-labels	分配标签到箱
POST	/api/packing-orders/:packingNumber/remove-labels	从箱移除标签
POST	/api/packing-orders/:packingNumber/regenerate-boxes	重新装箱
POST	/api/packing-orders/shipping-box-outbound	扫箱码出库
GET	/api/packing-orders/box-inventory	箱装库存查询

POST	<code>/api/packing-orders/:packingNumber/unpack</code>	整单拆箱
POST	<code>/api/packing-orders/unpack-boxes</code>	逐箱拆箱
GET	<code>/api/packing-orders/box/:boxNumber</code>	扫码查箱

19.7 报废处置 (scrap-disposal)

方法	路径	说明
GET	<code>/api/scrap-disposal/inventory</code>	报废仓库存
GET	<code>/api/scrap-disposal/inventory/:item_number</code>	批次明细
GET	<code>/api/scrap-disposal</code>	处置单列表
GET	<code>/api/scrap-disposal/:disposal_number</code>	处置单详情
POST	<code>/api/scrap-disposal</code>	创建处置申请
POST	<code>/api/scrap-disposal/:disposal_number/confirm</code>	确认处置(出库)
POST	<code>/api/scrap-disposal/:disposal_number/reject</code>	驳回
DELETE	<code>/api/scrap-disposal/:disposal_number</code>	删除
GET	<code>/api/scrap-disposal/monthly-report/completed-counts</code>	已完成盘点单
GET	<code>/api/scrap-disposal/monthly-report</code>	报废月度报表

19.8 其他

模块	方法	路径	说明
报废交易	GET	<code>/api/scrap-transactions</code>	报废仓流水查询
有效期报告	GET	<code>/api/shelf-life-report</code>	有效期管理报告

层级	文件	行数	核心职责
● 服务层	<code>server/src/services/warehouse/finishedGoods.ts</code>	1203	成品仓核心：生产入库、发货出库、退货入库、调整、撤回
● 服务层	<code>server/src/services/warehouse/material.ts</code>	631	物料仓核心：手工入库、半成品入库、手工出库、调整、撤回
● 服务层	<code>server/src/services/warehouse/helpers.ts</code>	252	通用函数： Upsert汇总、创建流水、FIFO扣减、批次追溯、会计期间校验
● 服务层	<code>server/src/services/warehouse/index.ts</code>	7	桶文件，统一导出

● 控制层	<code>modules/warehouse/finishedGoods/finishedGoods.controller.ts</code>	920	成品仓控制器：查询API+委托服务
● 控制层	<code>modules/warehouse/materialWarehouse/materialWarehouse.controller.ts</code>	768	物料仓控制器
● 控制层	<code>modules/warehouse/stockIn/stockIn.controller.ts</code>	865	采购入库控制器：CRUD+确认+撤回+对账
● 控制层	<code>modules/warehouse/stockIn/scrapInventoryHandler.ts</code>	344	报废入库库存处理：审批通过自动入库+反审回退
● 控制层	<code>modules/warehouse/stockCount/stockCount.controller.ts</code>	1269	盘点控制器：快照+录入+复核+确认+报表
● 控制层	<code>modules/warehouse/abnormalIO/abnormalIO.controller.ts</code>	1172	异常出入库：4种类型+确认+撤消反转
● 控制层	<code>modules/warehouse/packingOrder/packingOrder.controller.ts</code>	1165	装箱单：标签系统+自动装箱+确认+拆箱+扫箱码出库

● 控制层	<code>modules/warehouse/scrapDisposal/scrapDisposal.controller.ts</code>	618	报废处置：申请+确认出库+驳回
● 控制层	<code>modules/warehouse/scrapDisposal/scrapMonthlyReport.controller.ts</code>	165	报废仓月度报表
● 控制层	<code>modules/warehouse/scrapTransaction/scrapTransaction.controller.ts</code>	99	报废仓流水查询
● 控制层	<code>modules/warehouse/shelfLifeReport/shelfLifeReport.controller.ts</code>	110	有效期管理报告
● 控制层	<code>modules/warehouse/materialWarehouse/materialMonthlyReport.controller.ts</code>	369	原料仓月度报表
● 路由层	<code>modules/warehouse/index.ts</code>	26	仓储域路由聚合(9个子路由)
● 路由层	<code>modules/warehouse/*/*.routes.ts</code>	—	各子模块路由定义

 总计约 **8,400+** 行仓储相关代码，覆盖 **9个子模块**，**60+ API端点**

